

Experiment 02: Interfacing Dual Alternating LEDs with Arduino UNO

GPIO Digital Output Control — Program Source Code & Circuit State Output

1. Arduino C++ Source Code

Listing 2.1: Microcontroller Firmware for Periodic Anti-Phase LED Blinking.

```
1 /*
2  * Experiment 02: Dual Alternating LED Interfacing with Arduino UNO
3  * Hardware Pin Connections:
4  *   - Arduino Digital Pin 8 -> Green LED Anode (via 1 kOhm Resistor R1)
5  *   - Arduino Digital Pin 7 -> Red LED Anode (via 1 kOhm Resistor R2)
6  *   - Common Ground (GND) -> LED Cathodes
7  * Switching Interval: 1000 ms (1.0 second periodic period)
8  */
9 const int LED1_PIN = 8;    // Green LED Channel (Digital Pin D8)
10 const int LED2_PIN = 7;    // Red LED Channel (Digital Pin D7)
11 const unsigned long BLINK_INTERVAL = 1000; // Period: 1000 ms (1.0 s)
12
13 void setup() {
14   pinMode(LED1_PIN, OUTPUT); // Configure D8 as push-pull output driver
15   pinMode(LED2_PIN, OUTPUT); // Configure D7 as push-pull output driver
16   digitalWrite(LED1_PIN, LOW); // Initialize D8 LOW
17   digitalWrite(LED2_PIN, LOW); // Initialize D7 LOW
18 }
19
20 void loop() {
21   // Phase 1: LED1 ON (Active-HIGH 5V), LED2 OFF (Active-LOW 0V)
22   digitalWrite(LED1_PIN, HIGH);
23   digitalWrite(LED2_PIN, LOW);
24   delay(BLINK_INTERVAL);
25   // Phase 2: LED1 OFF (Active-LOW 0V), LED2 ON (Active-HIGH 5V)
26   digitalWrite(LED1_PIN, LOW);
27   digitalWrite(LED2_PIN, HIGH);
28   delay(BLINK_INTERVAL);
29 }
```

2. Experimental Output, Logic Timing & Truth Table

A. Circuit State Truth Table & Switching Log:

Interval	Pin 8 Output	Pin 7 Output	Observed Hardware State
0 – 1 s	HIGH (5 V)	LOW (0 V)	LED ₁ ON, LED ₂ OFF ($I_{F1} \approx 3$ mA, Green LED ON)
1 – 2 s	LOW (0 V)	HIGH (5 V)	LED ₁ OFF, LED ₂ ON ($I_{F2} \approx 3$ mA, Red LED ON)
2 – 3 s	HIGH (5 V)	LOW (0 V)	LED ₁ ON, LED ₂ OFF (Periodic Alternation)
3 – 4 s	LOW (0 V)	HIGH (5 V)	LED ₁ OFF, LED ₂ ON (Periodic Alternation)

B. Digital Timing Waveforms (Anti-Phase Switching, $T = 1.0$ s):

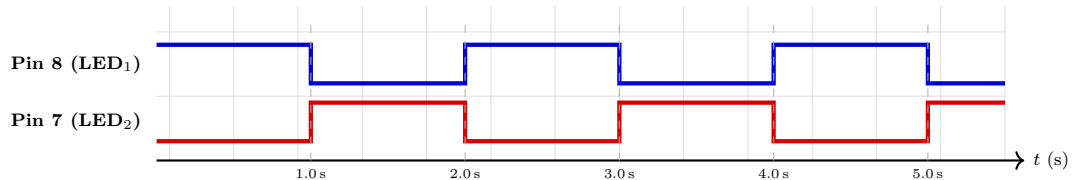


Figure 2.1: Digital logic anti-phase switching waveforms for complementary LED driving.